MILITARY SPECIFICATION SHEET

BOLTS, SHEAR, STEEL, 132 KSI Fsu, 450°F

This specification is approved for use by all Department and Agencies of the Department of Defense.

The complete requirements for procuring the bolts described herein shall consist of this document and the issue in effect of MIL-B-8906.

DESIGN INFORMATION:

The bolts covered by this specification sheet are for use where shear loads (not tensile) are the primary consideration.

The bolts are designed for use with nuts conforming to MIL-N-8985/2 and with washers conforming to MS21299.

Tooling shall be in accordance with MS33787 and MIL-W-8982.

THE ULTIMATE TENSILE STRENGTH, DOUBLE SHEAR STRENGTH, FATIGUE TEST LOAD AND MINIMUM HEAD BEARING AREA SHALL BE IN ACCORDANCE WITH TABLE III

REQUIREMENTS:

MATERIAL: ALLOY STEEL, AMS 6487

HARDNESS: ROCKWELL C46-C50

SURFACE TEXTURE: UNLESS OTHERWISE SPECIFIED SURFACE ROUGHNESS SHALL NOT EXCEED 125 MICROINCHES PER ANSI B46.1

PLATING: CADMIUM PLATE IN ACCORDANCE WITH MIL-C-8837

DESIGN AND CONSTRUCTION - SEE FIG. 1

FSC 5306

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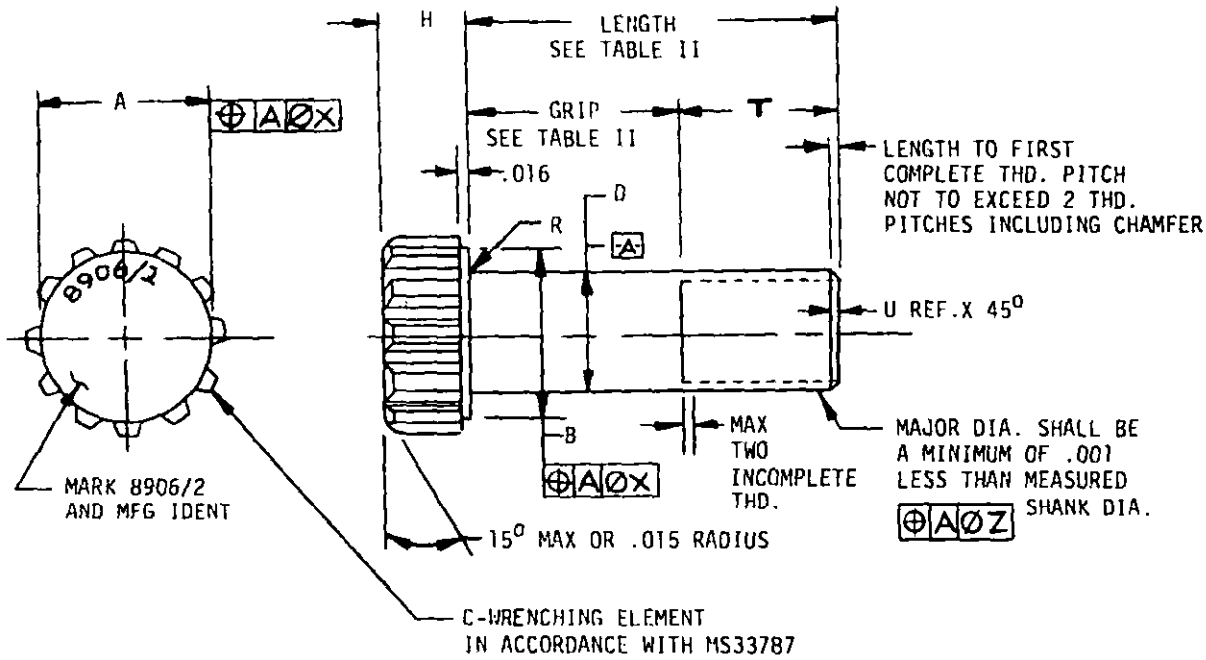


FIGURE 1 BOLT, SHEAR

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TABLE I BOLT SIZES

DASH NO.	THREAD UNJF-3A	A 1/ MIN.	B 1/ 4/		C WRENCHING ELEMENT	1/ 2/ 5/ D-DIA		H	R		T REF	U	CONCENTRICITY	
			MAX	MIN		MAX	MIN		MAX	MIN			X	Z
-3	10-32	.381	.380	.359	12	.1895	.1885	.140	.022	.016	.331	.04	.004	
-4	1/4-28	.447	.446	.422	14	.2495	.2485	.164	.028	.022	.397	.04	.005	.0045
-5	5/16-24	.512	.511	.484	16	.3120	.3110	.187	.035	.029	.467	.05	.006	
-6	3/8-24	.575	.574	.546	18	.3745	.3735	.210	.039	.038	.515	.05	.008	
-7	7/16-20	.707	.703	.671	22	.4370	.4360	.257	.049	.039	.599	.06	.009	
-8	1/2-20	.772	.784	.734	24	.4995	.4985	.281	.055	.045	.649	.06	.010	
-9	9/16-18	.901	.891	.859	28	.5615	.5605	.328	.061	.051	.725	.07	.011	.0060
-10	5/8-18	.967	.954	.922	30	.6240	.6230	.350	.068	.058	.777	.07	.012	
-12	3/4-16	1.096	1.063	1.031	34	.7490	.7480	.400	.080	.070	.914	.08	.015	
-14	7/8-14	1.291	1.250	1.218	40	.8740	.8730	.468	.093	.083	1.049	.09	.018	
-16	1-12	1.487	1.438	1.406	46	.9990	.9980	.539	.105	.095	1.218	.10	.020	.0090
-18	1 1/8-12	1.683	1.625	1.593	52	1.1240	1.1225	.609	.113	.108	1.334	.10	.022	
-20	1 1/4-12	1.813	1.812	1.781	56	1.2490	1.2475	.690	.130	.120	1.449	.10	.025	
-22	1 3/8-12	2.011	2.000	1.968	62	1.3740	1.3725	.750	.143	.133	1.576	.10	.028	.0120
-24	1 1/2-12	2.206	2.189	2.157	68	1.4990	1.4975	.820	.155	.145	1.700	.10	.030	

1/ DIMENSION "A", "B" AND "D" SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR "X" (see table I).

2/ DIMENSIONS "D" AND THREAD PITCH DIAMETER SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR "Z" T.I.R. (see table I).

3/ GRIP LENGTH OF BOLT SHALL BE MEASURED FROM UNDERSIDE OF HEAD TO THE END OF FULL CYLINDRICAL PORTION OF SHANK.

4/ "NATURAL FLOW OF MATERIAL PERMITTED TO A DIAMETER EQUAL TO B."

5/ ALL DIMENSION AFTER PLATING.

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TABLE II LENGTHS, DIAMETER AND DASH NUMBERS

DASH NO.	GRIP ±.010	LENGTH ±.015 DIAMETER DASH NUMBER													
		-3	-4	-5	-6	-7	-8	-9	-10	-12	-14	-16	-18	-20	-22	-24
		.1900	.2500	.3125	.3750	.4375	.5000	.5625	.6250	.7500	.8750	1.0000	1.1250	1.2500	1.3750	1.5000
-1	.062	.393	.459													
-2	.125	.456	.522	.592												
-3	.188	.519	.585	.655	.704	.787										
-4	.250	.581	.647	.717	.766	.849	.899									
-5	.312	.643	.709	.779	.828	.911	.961	1.037	1.089							
-6	.375	.706	.772	.842	.891	.974	1.024	1.100	1.152	1.289						
-7	.438	.769	.835	.905	.954	1.037	1.087	1.163	1.215	1.352	1.487	1.656				
-8	.500	.831	.897	.967	1.016	1.099	1.149	1.225	1.277	1.414	1.549	1.718	1.834			
-9	.562	.893	.959	1.029	1.078	1.161	1.211	1.287	1.339	1.476	1.611	1.780	1.896	2.011	2.138	
-10	.625	.956	1.022	1.092	1.141	1.224	1.274	1.350	1.402	1.539	1.674	1.843	1.959	2.074	2.201	2.325
-11	.688	1.019	1.085	1.155	1.204	1.287	1.337	1.413	1.465	1.602	1.737	1.906	2.022	2.137	2.264	2.388
-12	.750	1.081	1.147	1.217	1.266	1.349	1.399	1.475	1.527	1.664	1.799	1.968	2.084	2.199	2.326	2.450
-13	.812	1.143	1.209	1.279	1.328	1.411	1.461	1.537	1.589	1.726	1.861	2.030	2.146	2.261	2.388	2.512
-14	.875	1.206	1.272	1.342	1.391	1.474	1.524	1.600	1.652	1.789	1.924	2.093	2.209	2.324	2.451	2.575
-15	.938	1.269	1.335	1.405	1.454	1.537	1.587	1.663	1.715	1.852	1.987	2.156	2.272	2.387	2.514	2.638
-16	1.000	1.331	1.397	1.467	1.516	1.599	1.649	1.725	1.777	1.914	2.049	2.218	2.334	2.449	2.576	2.700
-17	1.062	1.393	1.459	1.529	1.578	1.661	1.711	1.787	1.839	1.976	2.111	2.280	2.396	2.511	2.638	2.762
-18	1.125	1.456	1.522	1.592	1.641	1.724	1.774	1.850	1.902	2.039	2.174	2.343	2.459	2.574	2.701	2.825
-19	1.188	1.519	1.585	1.655	1.704	1.787	1.837	1.913	1.965	2.102	2.237	2.406	2.522	2.637	2.764	2.888
-20	1.250	1.581	1.647	1.717	1.766	1.849	1.899	1.975	2.027	2.164	2.229	2.468	2.584	2.699	2.826	2.950
-21	1.312	1.643	1.709	1.779	1.828	1.911	1.961	2.037	2.089	2.226	2.361	2.530	2.646	2.761	2.888	3.012
-22	1.375	1.706	1.772	1.842	1.891	1.974	2.024	2.100	2.152	2.289	2.424	2.593	2.709	2.824	2.951	3.075
-23	1.438	1.769	1.835	1.905	1.954	2.037	2.087	2.163	2.215	2.352	2.487	2.656	2.772	2.887	3.014	3.138
-24	1.500	1.831	1.897	1.967	2.016	2.099	2.149	2.225	2.277	2.414	2.549	2.718	2.834	2.949	3.076	3.200
-25	1.562	2.893	1.959	2.029	2.078	2.161	2.211	2.287	2.339	2.476	2.611	2.780	2.896	3.011	3.138	3.262

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TABLE II (Continued)

DASH NO.	1/ GRIP ±.010	LENGTH ±.015 DIAMETER DASH NUMBER														
		-3	-4	-5	-6	-7	-8	-9	-10	-12	-14	-16	-18	-20	-22	-24
		.1900	.2500	.3125	.3750	.4375	.5000	5625	6750	.7500	.8750	1.0000	1.1250	1.2500	1.3750	1.5000
-26	1.625	1.956	2.022	2.092	2.141	2.224	2.274	2.350	2.402	2.539	2.674	2.843	2.959	3.074	3.201	3.325
-27	1.688	2.019	2.085	2.155	2.204	2.287	2.337	2.413	2.465	2.602	2.737	2.906	3.022	3.137	3.264	3.388
-28	1.750	2.081	2.147	2.217	2.266	2.349	2.399	2.475	2.527	2.664	2.799	2.968	3.084	3.199	3.326	3.450
-29	1.812	2.143	2.209	2.279	2.328	2.411	2.461	2.537	2.589	2.726	2.861	3.030	3.146	3.261	3.388	3.512
-30	1.875	2.206	2.272	2.342	2.391	2.474	2.524	2.600	2.652	2.789	2.924	3.093	3.209	3.324	3.451	3.575
-31	1.938	2.269	2.335	2.405	2.454	2.537	2.587	2.663	2.715	2.852	2.987	3.156	3.272	3.387	3.514	3.638
-32	2.000	2.331	2.397	2.467	2.516	2.599	2.649	2.725	2.777	2.914	3.049	3.218	3.334	3.449	3.576	3.700
-34	2.125	2.456	2.522	2.592	2.641	2.724	2.774	2.850	2.902	3.039	3.174	3.343	3.459	3.574	3.701	3.825
-36	2.250	2.581	2.647	2.717	2.766	2.849	2.899	2.975	3.027	3.164	3.299	3.468	3.584	3.699	3.826	3.950
-38	2.375	2.706	2.772	2.842	2.891	2.974	3.024	3.100	3.152	3.289	3.424	3.593	3.709	3.824	3.951	4.075
-40	2.500	2.831	2.897	2.967	3.016	3.099	3.149	3.225	3.277	3.414	3.549	3.718	3.834	3.949	4.076	4.200
-42	2.625	2.956	3.022	3.092	3.141	3.224	3.274	3.350	3.402	3.539	3.674	3.843	3.959	4.074	4.201	4.325
-44	2.750	3.081	3.147	3.217	3.266	3.349	3.399	3.475	3.527	3.664	3.799	3.968	4.084	4.199	4.326	4.450
-46	2.875	3.206	3.272	3.342	3.391	3.474	3.524	3.600	3.652	3.789	3.924	4.093	4.209	4.324	4.451	4.575
-48	3.000	3.331	3.397	3.467	3.516	3.599	3.649	3.725	3.777	3.914	4.049	4.218	4.334	4.449	4.576	4.700
-50	3.125	3.456	3.522	3.592	3.641	3.724	3.774	3.850	3.902	4.039	4.174	4.343	4.459	4.574	4.701	4.825
-52	3.250	3.581	3.647	3.717	3.766	3.849	3.899	3.975	4.027	4.164	4.299	4.468	4.584	4.699	4.826	4.950
-54	3.375	3.706	3.772	3.842	3.891	3.974	4.024	4.100	4.152	4.289	4.424	4.593	4.709	4.824	4.951	5.075
-56	3.500	3.831	3.897	3.967	4.016	4.099	4.149	4.225	4.277	4.414	4.549	4.718	4.834	4.949	5.076	5.200
-58	3.625	3.956	4.022	4.092	4.141	4.224	4.274	4.350	4.402	4.539	4.674	4.843	4.959	5.074	5.201	5.325
-60	3.750	4.081	4.147	4.217	4.266	4.349	4.399	4.475	4.527	4.664	4.799	4.968	5.084	5.199	5.326	5.450
-62	3.875	4.206	4.272	4.342	4.391	4.474	4.524	4.600	4.652	4.789	4.924	5.093	5.209	5.324	5.451	5.575
-64	4.000	4.331	4.397	4.467	4.516	4.599	4.649	4.725	4.777	4.914	5.049	5.218	5.334	5.449	5.567	5.700

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TABLE III BOLT STRENGTHS

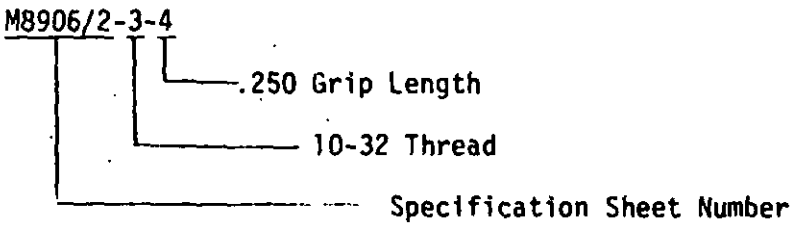
DIAMETER	ULTIMATE TENSILE STRENGTH LBS/MIN	DOUBLE SHEAR STRENGTH LBS/MIN	FATIGUE TEST LOAD LBS.		(1) HEAD BEARING AREA MIN. DIA. SQ. IN.
			HIGH LOAD	LOW LOAD	
-3	3,910	7,480	1,760	176	.0584
-4	6,980	13,000	3,140	314	.0666
-5	11,100	20,200	5,000	500	.0694
-6	17,100	29,100	7,700	770	.0733
-7	23,200	39,700	10,400	1,040	.1288
-8	30,900	51,800	13,900	1,390	.1314
-9	39,200	65,600	17,600	1,760	.2126
-10	49,000	81,000	22,000	2,200	.2140
-12	71,000	117,000	31,900	3,190	.1859
-14	97,100	159,000	43,700	4,370	.2827
-16	126,000	207,000	56,700	5,670	.4046
-18	162,000	262,000	72,900	7,290	.5617
-20	202,000	324,000	90,900	9,090	.7028
-22	247,000	392,000	111,000	11,100	.8776
-24	296,000	466,000	133,000	13,300	1.0840

(1) Design Information: Not a requirement for manufacturing or inspection.

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Part number: Consists of the basic number of this specification sheet and a dash number taken from Table II.

EXAMPLE:



"Revision Letters are not used to denote changes due to the extensiveness of the changes."

Custodians

ARMY-AV
NAVY-AS
AIR FORCE-11

Review Activities

AIR FORCE-99
DLA-IS

Preparing Activity

NAVY-AS

Project No. 5306-600

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-8906/2A BOLTS, SHEAR, STEEL, 132 KSI Fsu, 450°F

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE

2. REMARKS

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